

## \* Physics:

1. Presbyopia is the loss of ability of the eye to focus on near & far objects with advancing aging of a person.
2. Repulsion is the true test of charge.
3. virtual, erect, same size & laterally inverted.
4. It ~~se~~ should be aimed below the fishes image.

5. Given,

$$\text{Power} = 1.5 \text{ D}$$

$$f = \frac{1}{\text{Power}}$$

$$f = \frac{1}{1.5} \text{ m}$$

$$f = \frac{2}{3} \text{ m}$$

$$f = \frac{2}{3} \times 100 \text{ cm}$$

$$f = 66.7 \text{ cm}$$

$\therefore$  Focal length of the lens is 66.7 cm

$\therefore$  As the power is positive, the lens is a convex lens, meaning it is converging.

6. Given,

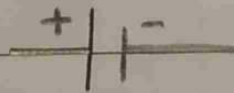
$$I = 5 \text{ A}$$

$$t = 30 \text{ mins} = 30 \times 60 = 1800 \text{ sec}$$

$$\begin{aligned} Q &= I \times t \\ &= 5 \text{ A} \times 1800 \text{ sec} \\ &= 9000 \text{ C} \end{aligned}$$

$\therefore$  A charge of 9000 C has circulated through the circuit in this time.

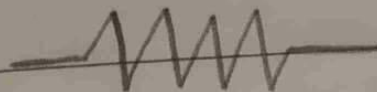
7. i) Electric Cell



ii) Switch in on position



iii) Resistor



iv) Electric Bulb



8.

Given,

$$h_1 = 5 \text{ cm}$$

$$u = -20 \text{ cm}$$

$$f = 12 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-20} = \frac{1}{12}$$

$$\frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{5 - 3}{60}$$

$$\frac{1}{v} = \frac{2}{60}$$

$$\frac{1}{v} = \frac{1}{30}$$

$$v = 30 \text{ cm}$$

$\therefore$  The image is formed 30 cm behind the convex lens; it is real & inverted

Height of image:

$$\frac{h_1}{h_2} = \frac{v}{u} = \frac{30}{-20} = -1.5$$

$\therefore$  Size of image 7.5 cm



9. Potential Difference  $\Rightarrow V = 10V - 5V$   
 $= 5V$

Work Done ( $W$ ) =  $5J$

Charge ( $Q$ ) = ?

We know that,

Potential Difference =  $\frac{\text{Work done}}{\text{Charge}}$

$$V = \frac{W}{Q}$$

$$Q = \frac{W}{V}$$

$$Q = \frac{5J}{5V}$$

$$Q = 1C$$

$\therefore$  The amount of charge flowing between the two ends is of  $1C$ .

10. i) Pole: It is the geometric centre of the reflecting surface. It is represented by the letter  $P$ .

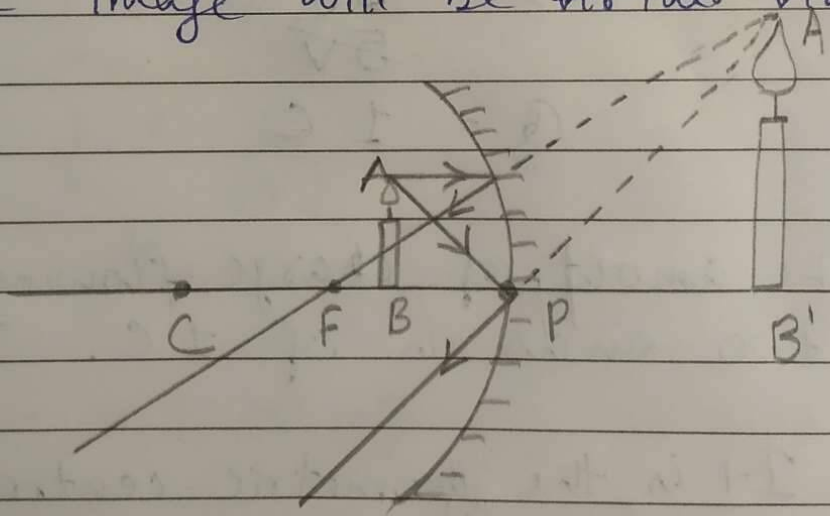
ii) Radius of Curvature: It is the radius of the sphere that forms the part of the mirror.

iii) Principal Axis: The straight line joining the

pole & the centre of curvature.

- iv) Principal focus: It is the point of the principal axis at which the rays parallel to the principal axis meet after reflection.

Object distance should be less than the focal length for the formation of an erect image, hence the range of distance of the candle flame should be less than 20 cm, i.e., between 0-20 cm. The image will be ~~virtule~~ virtual & erect.



- 11a) After the atmospheric scattering of light, only the red colour is able to reach our eyes because blue colour has a shorter wavelength & the red ~~col~~ colour has a longer wavelength. Thus, the sun appears reddish in the early morning.



- b) The ~~abitit~~ ability of the eye lens to adjust its focal length is called the power of accommodation of the eye.
- c) 25 cm is the least distance of distinct vision of a normal eye.